

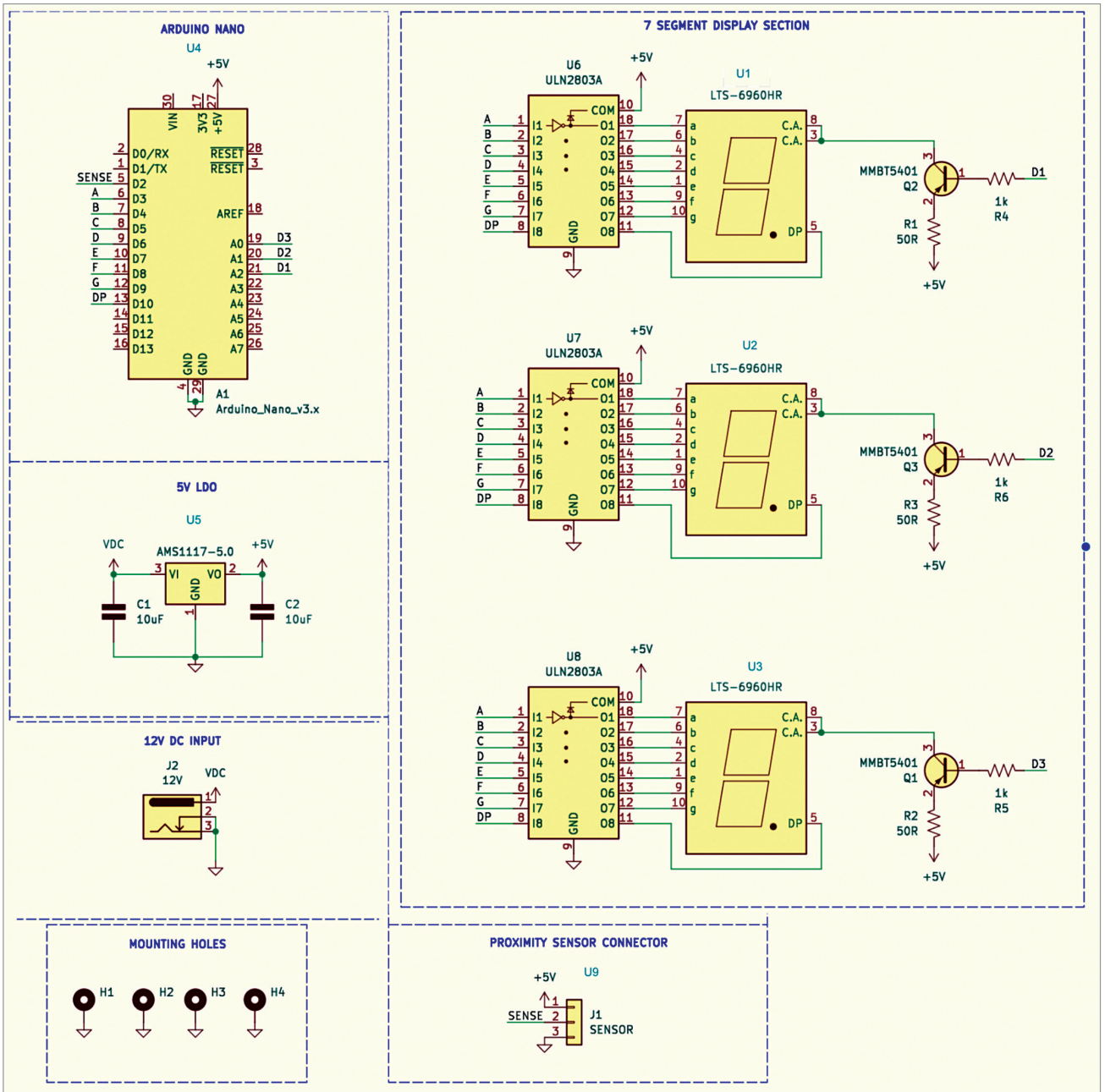


Fig. 3: Circuit diagram

all three appear continuously illuminated to the human eye.

The proximity sensor connects through header J1, which provides +5V, ground, and a signal line. When an object is detected, the sensor output changes state. Based on this input, the Arduino micro-controller processes the event and updates the numeric value displayed on the 7-segment modules.

The working of the circuit is

simple. When 12V DC power is applied, the Arduino power LED turns on, and the display shows '0', indicating that the system is ready. Each time a cement sack crosses the sensor, the count increases by one. The pushbutton on the Arduino Nano resets the count when required.

Software

The system is developed on the

Arduino platform, and the source code is written and compiled in the Arduino IDE. The SevSeg library is required to drive the 7-segment displays. Fig. 4 shows the procedure for installing the library.

Pins for the proximity sensor and the display are configured in the program. Interrupt pin D2 receives the sensor signal, while a combination of digital and analogue pins drives the segment and digit-select